

CyberVision

Multi-Source Cyber Threat Intelligence Integration and Visual Analytics Platform

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Github repo : [Cyber-Attack-Visualisation](#)

Abstract

In the contemporary digital landscape, organizations face an unprecedented volume of heterogeneous cyber threats, ranging from low-level network intrusions and malware memory injections to high-level geopolitical cyber warfare. Documenting and analyzing these threats requires the integration of disparate telemetry streams that, in most operational settings, are never brought into a single analytical frame. This report presents **CyberVision**, a modular, interactive, web-based visual analytics platform designed to unify global threat intelligence into one coherent, explorable system. CyberVision integrates five distinct datasets: global cybersecurity incident histories, state-sponsored cyber operations, software vulnerability records (CVEs), raw network intrusion signatures, and obfuscated malware memory dumps.

We implement a reproducible data cleaning and integration pipeline using `pandas` and Python, engineered so that every transformation — from column standardization to missing-value imputation — is deterministic and auditable. Geopolitical and industrial threats are evaluated via a dynamic, user-calibrated Country Risk Index ($\mathcal{R}$) computed using Min–Max normalized incident counts, average financial losses, and average operational resolution times, allowing analysts to reweight the index according to organizational priorities rather than relying on a single fixed notion of “risk.” The front end is developed as a Streamlit multi-page dashboard, featuring a custom glassmorphic dark-theme design system, custom CSS typography, and high-contrast neon-colored Plotly visualizations. Our findings indicate a strong correlation between highly digitized economies and increased threat exposure. Geopolitical threat analyses reveal nation-state actors prioritizing critical infrastructure and energy sectors, while malware memory diagnostics demonstrate a balanced class distribution suitable for downstream forensic models. This platform demonstrates the efficacy of interactive big data visualizations in aiding security analysts to synthesize complex logs and make data-driven defensive decisions.

Keywords: cyber threat intelligence, visual analytics, data integration, risk scoring, Streamlit, Plotly, malware forensics, geopolitical cyber operations.

Contents

Abstract	1
1 Introduction	4
1.1 Background	4
1.2 Problem Statement	4
1.3 Objectives	5
1.4 Scope	5
1.5 Motivation	5
1.6 Expected Outcomes	6
2 Dataset Description	7
2.1 Dataset Overviews	7
2.2 Data Dictionary and Schemas	7
2.3 Data Limitations and Boundaries	10
3 Methodology	12
3.1 Data Acquisition and Collection Pipeline	12
3.2 Data Cleaning and Standardization	13
3.3 Multi-Source Data Integration	13
3.4 Feature Engineering & Custom Risk Scoring	13
3.5 Frontend Layout, Styles, and UI Design	14
4 Exploratory Data Analysis	15
4.1 Dashboard Overview	15
4.2 Global Attack Volume Trend	15
4.3 Global Security Risk Map	16
4.4 Industry Impact Profile	17
4.5 Vulnerability Severity Distribution	18
4.6 Malware Family Analytics	19
4.7 Threat Attribution and Resolution Efficiency	20
4.8 Geopolitical Relationship Network	21
5 Results and Findings	23
5.1 Country Risk Leaderboard	23
5.2 Industry Sector Loss Projections	23
5.3 Network Traffic Protocol Share	23
5.4 Vulnerability Severity Breakdown	24
6 Discussion	25
6.1 Geopolitical and Economic Risk Dynamics	25
6.2 Vulnerability Lifecycle and Threat Landscape Shift	25
6.3 Practical Significance and Cyber Defense Tactics	25
6.4 Limitations of the Analysis	26

7	Challenges	27
7.1	High-Dimensional Volatility Memory Forensics	27
7.2	Data Inconsistencies and Missing Value Imputations	27
7.3	Network Graph Scaling and Layout Constraints	27
8	Conclusion	28
8.1	Summary of Accomplished Objectives	28
8.2	Key Analytical Insights	28
8.3	Platform Success and Practical Utility	28
9	Future Scope	29
9.1	Predictive Alerting Engines and Time-Series Forecasting	29
9.2	Integration of Standard CVSS Severity in Country Risk Calculations	29
9.3	Interactive Physics-Based Graph Renderings	29
9.4	Cloud Deployment and Real-Time Stream Integration	29
10	Code and Deployment Links	30
	References	31
A	Appendices	32
A.1	Appendix A: Complete Repository Folder Structure	32
A.2	Appendix B: Software and Package Requirements	32
A.3	Appendix C: Environment Setup and Execution Guide	33
A.4	Appendix D: Important Preprocessing & Risk Logic Snippets	33

Chapter 1

Introduction

1.1 Background

The digital revolution has transformed standard economic, industrial, and social activities. Cloud infrastructures, distributed database clusters, and interconnected networks are now standard operational foundations for organizations of every size, from small enterprises to national utilities. This expansion of the digital footprint has, however, significantly enlarged the vulnerability surface available to adversarial entities. Every new endpoint, API, or third-party integration represents a potential ingress point for an attacker, and the rate at which these surfaces are added routinely outpaces the rate at which they can be secured.

Security operations centers (SOCs) are constantly bombarded with telemetry logs generated across multiple network endpoints. These logs are highly heterogeneous, ranging from packet-level capture logs and memory forensic states to high-level incidents of industrial disruption and attribution metadata. Making sense of this overwhelming, high-velocity data requires robust data engineering pipelines and intuitive big data visualizations. Without visual synthesis, security analysts suffer from “alert fatigue” and struggle to find correlations across different threat layers — a network anomaly, a newly disclosed CVE, and a geopolitical incident report may all describe facets of the same emerging campaign, yet in most SOC tooling they are never viewed side by side.

1.2 Problem Statement

Modern cyber threat intelligence (CTI) suffers from severe fragmentation. A security analyst trying to assess organizational risk might have to inspect:

- Network traffic capture logs to find ongoing intrusions or port scans.
- Volatility-based memory dumps to detect obfuscated malware processes running on local servers.
- The National Vulnerability Database (NVD) to identify software components requiring emergency patches.
- Geopolitical news feeds to understand active Threat Actors targeting their specific industry sector.

Each of these data sources resides in its own isolated silo, using distinct formats, timestamps, schemas, and standards. The lack of standard tools to ingest, standardize, and visually integrate these disparate layers means that organizations struggle to build a unified defensive posture. An analyst who wants to answer a simple question — “is our exposure this quarter higher than last quarter, and why?” — is instead forced to manually reconcile spreadsheets, dashboards, and PDF advisories from unrelated systems. Bridging this gap requires a platform that standardizes these telemetry feeds, integrates them across spatial (geographic) and temporal (yearly) dimensions, and presents them in a single, high-fidelity visual dashboard.

1.3 Objectives

The primary objective of this project is to develop and document **CyberVision**, an interactive visual analytics dashboard designed to address the fragmentation of threat intelligence. The specific technical goals include:

1. **Pipeline Design:** Build a reproducible, modular Python pipeline to automate data downloading, cleansing, and schema standardization for five distinct CTI datasets.
2. **Schema Integration:** Define mapping rules to standardize geographical codes, threat category naming, and temporal timestamps to support multi-dataset queries.
3. **Risk Score Formulation:** Implement a dynamic, parameterized Country and Industry Risk Index ($\mathcal{R}$) using Min-Max scaling that allows analysts to adjust feature weights (Frequency, Economic Loss, Resolution Time) based on organizational priorities.
4. **Visual Analytics:** Implement high-contrast, interactive visualizations (choropleth maps, treemaps, heatmaps, scatter forensics, and relation networks) to expose hidden threat patterns.
5. **Premium Interface:** Build a Streamlit multi-page web application featuring responsive grid cards, glassmorphic dark-theme styling, and cached file loader routines to support rapid analytics exploration.

1.4 Scope

The scope of CyberVision is defined by the ingestion and analysis of five core datasets representing different levels of the security stack:

- **Macroscopic Level:** Global cybersecurity threats (2015–2024) and the Council on Foreign Relations (CFR) database of state-sponsored cyber operations (2005–2020).
- **Vulnerability Level:** GitHub Security Advisory and CVE registry datasets (2021–2023) mapping software vulnerabilities.
- **Microscopic Traffic Level:** Packet-level network intrusion signatures (~40,000 traffic logs) detailing protocols and anomalies.
- **Forensic Memory Level:** Obfuscated memory signatures from the CIC-MalMem-2022 dataset (~58,000 records) mapping Trojan, Ransomware, and Spyware behavior.

The geographical focus covers ten standardized industrial economies (Australia, Brazil, China, France, Germany, India, Japan, Russia, United Kingdom, and the United States). Restricting the analysis to this fixed set of economies keeps the geospatial and comparative visualizations tractable while still spanning a representative mix of highly digitized, emerging, and state-controlled digital markets. The application is hosted as a single-node Python Streamlit service, caching processed data locally for efficient retrieval, which keeps the deployment footprint minimal while the platform is in its analytical, pre-production stage.

1.5 Motivation

This project is motivated by the critical need to support human analysts in CTI synthesis. Standard security tools (SIEMs) present logs in tabular grids that mask trends; a table of ten thousand rows can encode a clear seasonal spike or a shift in attacker tooling, but that pattern

is invisible until it is plotted. By applying data science methodologies and visualization best practices, we transform rows of logs into interactive, spatial-temporal dashboards that surface such patterns directly.

Furthermore, the inclusion of a customizable risk index is motivated by the realization that “risk” is not a static property; a financial institution cares more about financial loss, whereas a public utility company prioritizes operational downtime. A single, hard-coded risk formula would therefore misrepresent risk for at least some class of organization. CyberVision empowers analysts to align the analytics to their specific risk criteria by exposing the underlying weights as first-class, adjustable parameters rather than burying them in code.

1.6 Expected Outcomes

We expect the implementation of CyberVision to produce:

- A standardized, cleaned data repository derived from raw, noisy, and missing-laden CSV source files.
- A fully interactive dashboard running locally or on local intranets.
- Direct insights into the peak periods of cyber activity, the targeting profiles of nation-state actors, the correlation of network anomaly scores with payload sizes, and the features that distinguish benign memory signatures from malware.
- A clear blueprint for future extensions, including machine learning-based forecasting and live stream logging.

Chapter 2

Dataset Description

2.1 Dataset Overviews

CyberVision integrates five distinct datasets to provide a multi-layered perspective of cyber threats, moving from strategic, nation-level phenomena down to raw, packet-level and memory-level forensic signals. These datasets are described below.

1. **Global Cybersecurity Threats (2015–2024)**: This dataset contains historical records of macroscopic cyber attacks. It contains variables detailing the country of target, the year of incident, the category of attack, the target commercial sector, economic damage (financial loss), the number of affected user accounts, the threat source category, the vulnerability type exploited, the primary defense mechanism deployed, and the operational time taken to resolve the incident.
2. **Council on Foreign Relations (CFR) Cyber Incidents (2005–2020)**: Focuses on high-level, state-sponsored cyber operations. It includes records of espionage, sabotage, and theft attributed to nation-states. It contains metadata on the attacker affiliations, sponsor countries, victim organizations, categories of target, and the official responses.
3. **Security Vulnerabilities (CVE)**: Formed of software vulnerability advisories published between 2021 and 2023. It maps specific CVE numbers to their severity levels (Critical, High, Moderate, Low), publishing timestamps, text descriptions of the vulnerability vector, and references.
4. **Network Intrusion Signatures**: A dataset containing packet-level network traffic forensics. It maps traffic events characterized by source/destination IPs, ports, transport protocols (TCP, UDP, ICMP), packet lengths, packet classes (Control, Data), traffic applications (DNS, HTTP, FTP), and numerical anomaly scores.
5. **CIC-MalMem-2022 (Malware Memory Forensics)**: An obfuscated memory dataset from the Canadian Institute for Cybersecurity. It contains Volatility-extracted features from memory dumps of systems infected by Trojan, Spyware, or Ransomware, alongside benign memory snapshots. It contains 58 variables representing active processes, active DLLs, open handles, API hook callbacks, and registry keys.

Together, these five sources span roughly two decades of activity and cover strategic, tactical, and forensic granularities, which is precisely the range of resolution a working SOC analyst must reconcile on a daily basis.

2.2 Data Dictionary and Schemas

Below are the structured data dictionaries for the processed datasets, listing each retained column, its inferred data type after cleaning, a description, the observed missing-value ratio, and representative sample values or distributions.

Table 2.1: Global Cybersecurity Threats Schema (global_threats_clean.csv)

Column	Type	Description	Missing	Sample Values / Distribution
country	Object	Country where the target entity resides.	0.0%	“UK”, “Brazil”, “India”, “Japan”, “France”
year	Int64	Year of the incident (2015–2024).	0.0%	Mean: 2019.57, Range: 2015–2024
attack_type	Object	Standardized classification of the attack.	0.0%	“DDoS” (531), “Phishing” (529), “SQL Injection” (503)
target_industry	Object	Commercial/public sector targeted.	0.0%	“IT” (478), “Banking” (445), “Healthcare” (429)
financial_loss_in_million_	Float64	Financial damage in millions of USD.	0.0%	Mean: \$50.49M, Range: \$0.5M–\$99.99M
number_of_affected_users	Int64	Number of user accounts compromised.	0.0%	Mean: 504,684, Range: 424–999,635
attack_source	Object	Categorical description of the threat actor.	0.0%	“Nation-state” (794), “Insider” (752), “Hacker Group”
security_vulnerability_type	Object	Structural vulnerability exploited.	0.0%	“Zero-day” (785), “Social Engineering” (747)
defense_mechanism_used	Object	Primary security defense configuration.	0.0%	“Antivirus” (628), “VPN” (612), “Encryption” (592)
incident_resolution_time_in_hours	Int64	Total hours taken to remediate the attack.	0.0%	Mean: 36.48 hrs, Range: 1–72 hrs

Table 2.2: CFR Cyber Operations Incidents Schema (cfr_incidents_clean.csv)

Column	Type	Description	Missing	Sample Values / Distribution
title	Object	High-level summary title of the operation.	0.0%	“Attack on Austrian foreign ministry”
date	Object	Calendar date of discovery.	0.0%	“2020-02-13” (parsed to date-time)
affiliations	Object	Suspected state-sponsored hacker group.	0.0%	“Turla”, “APT28”, “Unknown”
description	Object	Paragraph describing the breach context.	0.0%	“Suspected Russian hackers conducted attacks...”
response	Object	Official victim response or confirmation.	0.0%	“Confirmation http://...”
victims	Object	Specific target organization or agency.	0.0%	“Austrian Foreign Ministry”
sponsor	Object	Suspected state sponsoring the attack.	0.0%	“Russian Federation”, “China”, “Iran”
type	Object	Goal of the cyber operation.	0.0%	“Espionage”, “Sabotage”, “Theft”
category	Object	General sector of target organization.	0.0%	“Government”, “Military”, “Energy”
year	Float64	Extracted year of incident.	0.0%	Range: 2005–2020

Table 2.3: Security Vulnerabilities Schema (vulnerabilities_clean.csv)

Column	Type	Description	Missing	Sample Values / Distribution
title	Object	Advisory headline or vulnerability title.	0.0%	“Ghost vulnerable to arbitrary file read...”
date	Object	Timestamp of publication.	0.0%	“2023-08-15 20:35:20+00:00”
severity	Object	Severity rating of the vulnerability.	0.0%	“Moderate” (1447), “High” (1048), “Critical” (449)
summary	Object	Advisory details indicating the package.	0.0%	“CVE-2023-40028 published for ghost(npm)”
link	Object	URL linking to GitHub advisory registry.	0.0%	“https://github.com/advisories/GHSA-9c9v...”
year	Int64	Publication calendar year.	0.0%	Range: 2021–2023

Table 2.4: Network Intrusion Signatures Schema (attack_signatures_clean.csv)

Column	Type	Description	Missing	Sample Values / Distribution
timestamp	Object	Date and time of network packet log.	0.0%	“2023-05-30 06:33:58”
source_ip_address	Object	IP address where packet originated.	0.0%	“103.216.15.12”
destination_ip_address	Object	IP address of target system.	0.0%	“84.9.164.252”
source_port	Int64	Port of origin.	0.0%	Mean: 32,970, Range: 1027–65530
destination_port	Int64	Port targeted on destination.	0.0%	Mean: 33,150, Range: 1024–65535
protocol	Object	Transport layer protocol.	0.0%	“ICMP” (13429), “UDP” (13299), “TCP” (13272)
packet_length	Int64	Length of packet payload in bytes.	0.0%	Mean: 781.45 bytes, Range: 64–1500
packet_type	Object	Traffic classification.	0.0%	“Control” (20237), “Data” (19763)
traffic_type	Object	Application layer protocol.	0.0%	“DNS” (13376), “HTTP” (13360), “FTP” (13264)
payload_data	Object	Obfuscated alphanumeric payload string.	0.0%	“Qui natus odio asperiores nam...”

Table 2.5: CIC-MalMem-2022 Forensic Memory Schema (malmem_clean.csv)

Column	Type	Description	Missing	Sample Values / Distribution
category	Object	Broad class classification.	0.0%	“Benign” (29231), “Spyware” (9815), “Ransomware” (9529)
class	Object	Binary class label.	0.0%	“Benign” (29231), “Malware” (28831)
family	Object	Specific malware family identifier.	0.0%	“Benign” (29231), “Transponder” (2410), “Shade” (2128)
pslist_nproc	Int64	Number of active processes.	0.0%	Mean: 41.4, Range: 21–240
dlllist_ndlls	Int64	Total DLLs loaded in memory.	0.0%	Mean: 1810, Range: 144–4500
handles_nhandles	Int64	Total open handles.	0.0%	Mean: 9200, Range: 800–15000
malfind_ninjections	Int64	Count of detected malicious injections.	0.0%	Mean: 5.2, Range: 0–45
callbacks_ncallbacks	Int64	Total registered kernel callbacks.	0.0%	Mean: 87.2, Range: 40–120

2.3 Data Limitations and Boundaries

Every data analytical framework contains structural limitations that must be addressed to preserve analytical integrity. In CyberVision, three limitations are especially important to keep in mind when interpreting any cross-dataset visualization:

- **Temporal Non-Alignment:** The datasets span different temporal ranges. The CFR Cyber Operations dataset terminates in 2020, while the Security Vulnerability (CVE) dataset only commences in 2021. The Global Cybersecurity Threats dataset spans 2015–2024. Consequently, integrated queries across all variables can only be conducted on overlapping year windows (specifically 2015–2020 for threats + geopolitical operations, and 2021–2024 for threats + vulnerabilities). Any trend line that appears to span the full 2005–2024 range is therefore an overlay of sub-ranges rather than a single continuously observed series.
- **Hardcoded Geographical Scopes:** The spatial integration engine aligns geographical data by mapping text-based attributes to ten standard industrial nations. Country names outside this defined boundary are title-cased using fallback routines, meaning the platform is intentionally not exhaustive of every country represented in the raw sources.
- **Malware Imbalances:** Although the binary classes (Benign vs. Malware) are balanced, specific sub-families of Trojan (e.g. Zeus vs. Tibs) exhibit imbalances that preclude sub-family visualization. Thus, the system groups malware by top-level categories rather than attempting fine-grained family-level breakdowns that would be statistically unreliable.

Chapter 3

Methodology

The implementation workflow of CyberVision follows a modular, pipeline-oriented structure designed to ensure clean separation of concerns and reproducibility. Raw data is never touched directly by the presentation layer; it instead passes through a strictly ordered sequence of pre-processing, integration, and backend logic stages before being rendered, which makes each stage independently testable and re-runnable.

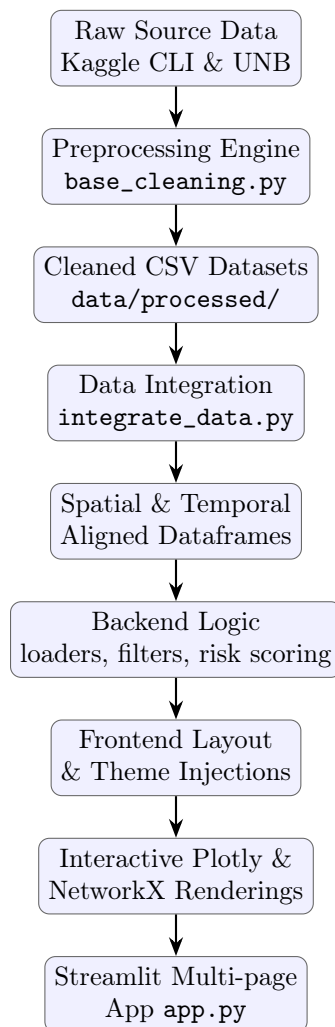


Figure 3.1: End-to-end CyberVision data and rendering pipeline.

3.1 Data Acquisition and Collection Pipeline

Raw datasets are acquired from Kaggle and host portals. An automation script (`src/preprocessing/download_`) uses the Kaggle API CLI to pull raw zip archives directly into the `data/raw/` directory and extract them programmatically, removing the need for any manual downloading step for four of the

five datasets. The CIC-MalMem-2022 dataset requires manual download from the UNB dataset portal due to access controls, and is placed in `data/raw/cic_malmem_2022.csv`.

3.2 Data Cleaning and Standardization

Standardization and data cleansing are handled by `src/preprocessing/base_cleaning.py`. This module provides reusable functions imported by dataset-specific cleaning scripts, so that every dataset is subject to the same baseline quality guarantees before it is allowed into the processed layer:

- **Column Standardization:** Column headers are cleaned to prevent query syntax errors, using the transformation

Cleaned Header = lowercase (strip(whitespace → “_” and special chars → “”).

- **Duplicate Removal:** Entire row duplicates are detected and removed based on unique combinations (e.g., `title` and `date` for vulnerabilities).
- **Date Standardization:** Text-based timestamps are coerced into pandas datetime objects using `errors="coerce"` to handle corrupted strings. The calendar year is extracted as a separate analytical category.
- **Categorical Trimming:** Text categoricals (e.g. ‘USA’, ‘usa’, ‘Usa ’) are stripped of trailing spaces and title-cased to resolve category fragmentation during aggregations.
- **Imputation Strategy:** Missing numeric values are filled using median values to preserve the skew of network traffic metrics. Categorical columns are filled with “Unknown” (or “Unrated” for CVE severities) to avoid dropping records.

3.3 Multi-Source Data Integration

Integration occurs along two primary axes:

1. **Spatial Integration:** Handled by `integrate_spatial()`. It aggregates the global threats dataset by standard country, summing incident counts, financial losses, affected user volumes, and average operational resolution hours.
2. **Temporal Integration:** Handled by `integrate_temporal()`. It groups threats, state-sponsored CFR incidents, and CVE disclosures by calendar year, executing outer joins to build a comprehensive historical trend timeline (2005–2024).

These two integration axes are what allow the same underlying cleaned tables to feed both the geospatial views (Chapter 4) and the year-over-year trend views without duplicating logic across pages.

3.4 Feature Engineering & Custom Risk Scoring

To evaluate the threat profile of different entities, we develop a custom, parameterized Risk Index ($\mathcal{R}$) computed out of 10.0. The risk score for an entity (country or industry) is calculated as a weighted sum of three Min–Max normalized metrics:

1. **Incident Count (N):** The frequency of attacks, representing target popularity.
2. **Average Financial Loss (L):** The economic damage impact.
3. **Average Resolution Time (T):** The operational difficulty of remediation.

Normalization Formulation

For each raw metric vector $\mathbf{X} = \{x_1, x_2, \dots, x_n\}$, the normalized value $\bar{x}_i \in [0, 1]$ is computed as:

$$\bar{x}_i = \frac{x_i - \min(\mathbf{X})}{\max(\mathbf{X}) - \min(\mathbf{X})}.$$

If the maximum and minimum are equal, the system assigns a default scale of 0.5 to prevent division-by-zero errors, which occurs when an entity is the only member of its group and therefore has no internal spread to normalize against.

Weighted Formulation

The risk score $\mathcal{R}_i \in [0.0, 10.0]$ is computed as:

$$\mathcal{R}_i = (w_c \cdot \bar{N}_i + w_l \cdot \bar{L}_i + w_t \cdot \bar{T}_i) \times 10,$$

where:

- w_c is the weight of incident count (default: 0.3),
- w_l is the weight of financial loss (default: 0.4),
- w_t is the weight of resolution time (default: 0.3),
- subject to the constraints $w_c + w_l + w_t = 1.0$ and $w_c, w_l, w_t \geq 0.0$.

The sidebar UI provides sliders allowing analysts to calibrate these weights in real time, which recalculates the risk leaderboard across all tabs. Because the weights are constrained to sum to one, the resulting score remains bounded within $[0, 10]$ regardless of how an analyst redistributes emphasis across the three components, which keeps the leaderboard visually comparable across different weighting regimes.

3.5 Frontend Layout, Styles, and UI Design

The UI of CyberVision is designed to provide a premium, modern, glassmorphic look that distinguishes it from the default styling of most Streamlit dashboards.

- **Glassmorphism Theme:** Injected via custom HTML/CSS stylesheet wrappers (`frontend/layout.py`). Grid containers have a translucent background (`rgba(17, 25, 40, 0.65)`) with high saturation blur (`backdrop-filter: blur(16px)`), subtle borders (`rgba(255, 255, 255, 0.085)`), and neon hover highlights.
- **Custom Typography:** We import Google Fonts (**Inter** for standard labels, **Outfit** for numerical headers and dashboard titles).
- **Neon Color Palette:** Plotly charts are styled with specific colors: Neon Cyan (`#00f2fe`) for primary data series, Neon Pink (`#ff0055`) for risk indicators, and Neon Green (`#00ff87`) for normal status variables.
- **Caching:** Performance is enhanced using Streamlit's `@st.cache_data` loader to prevent disk read bottlenecks during widget updates, so that repeated interactions with sliders and filters do not re-trigger a full disk read of the processed datasets.

Chapter 4

Exploratory Data Analysis

Each of the following sub-sections documents one visualization implemented in CyberVision: its analytical purpose, what it visually represents, the observations drawn from it, its interpretation in a business/security context, and the key takeaway an analyst should walk away with.

4.1 Dashboard Overview

Before drilling into individual charts, it is useful to describe the overall shell that every page of CyberVision shares, since this shell is what gives the platform its at-a-glance, single-pane-of-glass character.

- **Global Filters & Risk Index Calibration:** The persistent left sidebar exposes three linked sliders — Frequency Weight, Financial Impact Weight, and Resolution Time Weight (defaulting to 0.30, 0.40, and 0.30 respectively) — which feed directly into the Risk Index formula of Section 3.5 and recompute every leaderboard and choropleth on the page without a full reload.
- **KPI Summary Cards:** Four glassmorphic metric cards sit at the top of every page: Total Global Incidents (3,000), Estimated Loss in USD (\$151,478.9M), Users Impacted (1,514,052,409), and Average Resolution Time (36.5 hrs). These act as a constant, page-independent point of reference while an analyst navigates between views.
- **Tab-Based Navigation:** A horizontal tab bar switches between the seven analytical views covered in this chapter — Country Threat Map, Global Threat Trends, Industry Risk Dashboard, Vulnerability Explorer, Malware Family Analytics, Attack Source & Resolution, and Threat Relationship Network — without triggering a full page reload, since the underlying dataframes are cached once via `@st.cache_data` and simply re-filtered per tab.
- **Export Datasets:** Sidebar buttons let an analyst download the currently filtered spatial risk data and temporal trends data as CSV, so that findings surfaced interactively in the dashboard can be taken into offline reporting tools.
- **GitHub Dataset Registry:** A compact sidebar panel lists the row counts of each underlying source table — Global Threat Logs (3,000 rows), CFR Incidents Catalog (28 rows), Vulnerability Registry (3,111 rows), and Malware Forensic Dumps (58,062 rows) — giving analysts immediate visibility into the scale and provenance of the data backing whichever chart they are viewing.

This consistent shell is what allows the seven page-specific views documented below to be read as facets of one coherent investigation rather than as disconnected reports.

4.2 Global Attack Volume Trend

- **Purpose:** To observe the historical growth and trajectory of global cyber attacks.

- **What it represents:** An interactive area plot charting the total incident count per year.
- **Observations:** The chart shows a minimal attack frequency from 2005 to 2014, reflecting the boundary of the CFR dataset. From 2015 onwards, when the Global Threats dataset is integrated, incident volumes rise, peaking in 2017 (319 incidents) and 2022 (318 incidents), before stabilizing around 299 in 2024.
- **Business Interpretation:** The surges correspond to the emergence of global ransomware crises (e.g. WannaCry, NotPetya in 2017) and heightened geopolitical tensions in 2022.
- **Key Takeaway:** Cyber risk is highly dynamic; organizations must scale their budgets in response to macro-level trends.

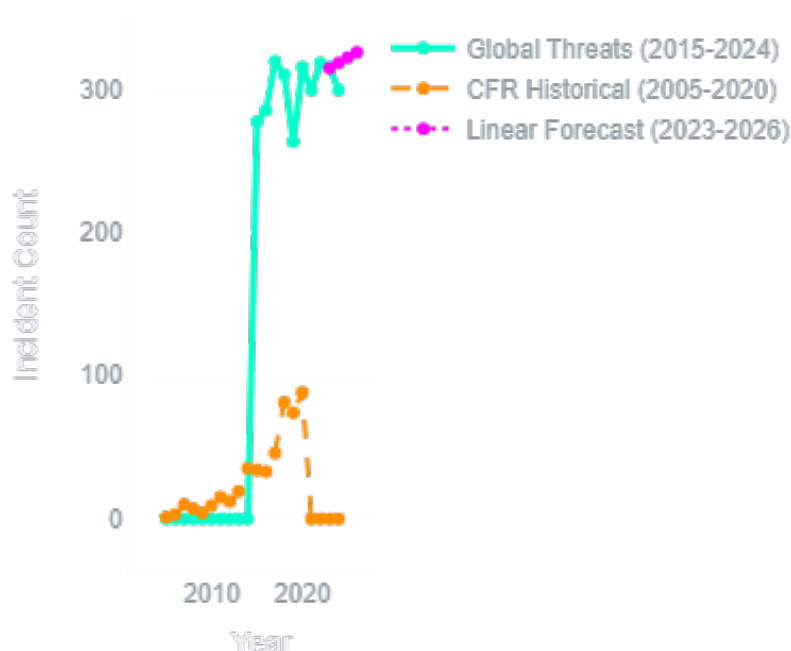


Figure 4.1: Chronological cyber incident trend timeline with a 3-year regression forecast.

4.3 Global Security Risk Map

- **Purpose:** To analyze the geopolitical distribution of cyber risk.
- **What it represents:** A world choropleth map colored by the dynamic Country Risk Index.
- **Observations:** With standard weights (Count: 0.3, Loss: 0.4, Time: 0.3), the United Kingdom and Brazil register the highest risk indexes (7.68 and 6.97, respectively), while Germany (5.58) and Japan (5.31) show moderate-to-high risk. In contrast, China (2.57) shows a low risk score.
- **Business Interpretation:** Highly digitized, service-oriented economies face greater exposure to financial losses and report incidents more frequently, while less integrated or highly controlled networks exhibit lower visible incident profiles.
- **Key Takeaway:** Cyber risk does not map simply to geographical size; it correlates with digital asset concentration and local regulatory disclosure laws.

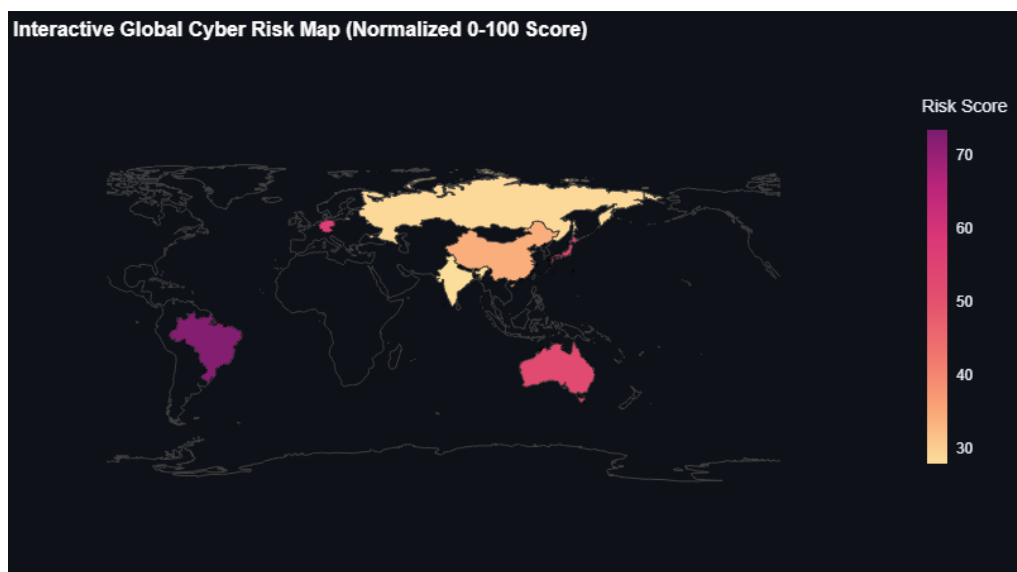


Figure 4.2: Interactive world risk map rendered under the dynamic calibration weights.

4.4 Industry Impact Profile

- **Purpose:** To evaluate sector-specific vulnerabilities.
- **What it represents:** A treemap where block size represents total financial loss (\$M) and color indicates average loss per incident.
- **Treemap Encoding:** The treemap utilizes a dual-variable encoding scheme:
 - *Block Size:* Encodes the cumulative (total) financial loss of the sector or vector, showing which categories represent the largest absolute financial drains.
 - *Color Scale:* Encodes the financial loss value range (from dark teal for low loss to bright cyan for maximum loss), highlighting the intensity of individual threat vectors.
- **Observations:** The IT, Healthcare, and Finance sectors occupy the largest sections. Healthcare has a high total financial loss, while IT exhibits the highest average financial damage per breach.
- **Critical Sectors:** Sectors like Healthcare and Finance typically show large dimensions due to substantial breach response costs (e.g. data restoration and regulatory penalties).
- **Business Interpretation:** Healthcare organizations incur high recovery costs due to compliance penalties and system restoration, while IT infrastructure targets represent direct compromise paths for software supply chain exploits.
- **Weight Calibrator:** Changing the sidebar weights (Frequency: 30%, Loss: 40%, Operational: 30%) shifts the Risk Index values in the leaderboard in real time, letting an analyst immediately see how the sector ranking would change under a different risk priority.
- **Key Takeaway:** Zero-trust policies must be customized; financial sectors require immediate transaction monitoring, whereas healthcare requires robust data recovery capabilities.



Figure 4.3: Treemap profile of financial loss by industry sector.

4.5 Vulnerability Severity Distribution

- **Purpose:** To track the distribution of software vulnerabilities across severity tiers.
- **What it represents:** An interactive donut (annular pie) chart segmenting the full population of published CVEs into four severity classes — Moderate, High, Critical, and Low — with each wedge's arc length proportional to its share of total disclosures.
- **Observations:** Moderate-severity CVEs dominate the distribution at 46.5%, followed by High severity at 33.7%. Critical-severity disclosures account for 14.4% of the total, while Low-severity CVEs form the smallest slice at 5.4%. Together, High and Critical severities constitute nearly half (48.1%) of all disclosed vulnerabilities.
- **Business Interpretation:** The rapid adoption of cloud microservices has expanded the attack surface, leading to a substantial share of High- and Critical-rated vulnerabilities. A distribution this heavily weighted toward Moderate-and-above severities indicates that a “patch everything eventually” posture is insufficient; a large fraction of the disclosure volume genuinely warrants urgent attention.
- **Key Takeaway:** Vulnerability patch management must move away from “patch-everything” strategies to focus on critical assets, using the severity donut as a quick, at-a-glance triage signal before drilling into the underlying CVE list.

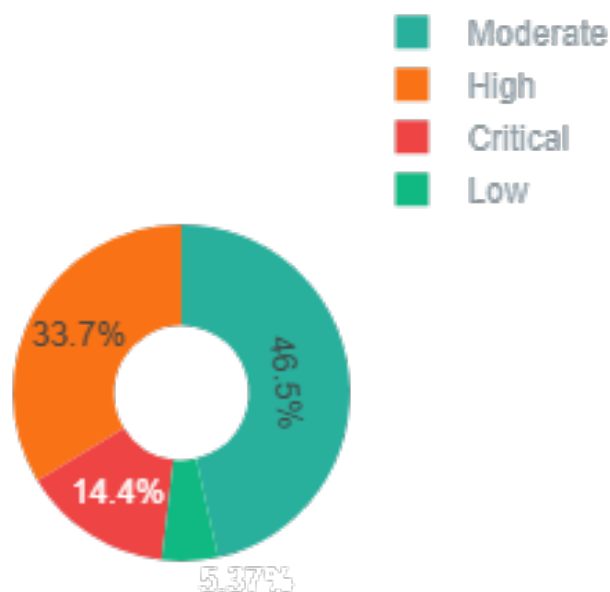


Figure 4.4: Vulnerability severity distribution donut chart: Moderate (46.5%), High (33.7%), Critical (14.4%), Low (5.4%).

4.6 Malware Family Analytics

- **Purpose:** To analyze memory-level forensic categories.
- **What it represents:** A bar chart mapping the distribution of malware families (Trojan, Spyware, Ransomware) against Benign records.
- **Observations:** The class breakdown is balanced: Benign (29,231), Spyware (9,815), Ransomware (9,529), and Trojan (9,487). The top-performing malware families are Transponder (2,410) and Shade (2,128).
- **Business Interpretation:** Obfuscated memory analysis requires class-balanced forensic models to prevent high false-positive rates during endpoint detection.
- **Key Takeaway:** Memory forensics is key for detecting modern obfuscated threats that bypass signature-based antivirus tools.

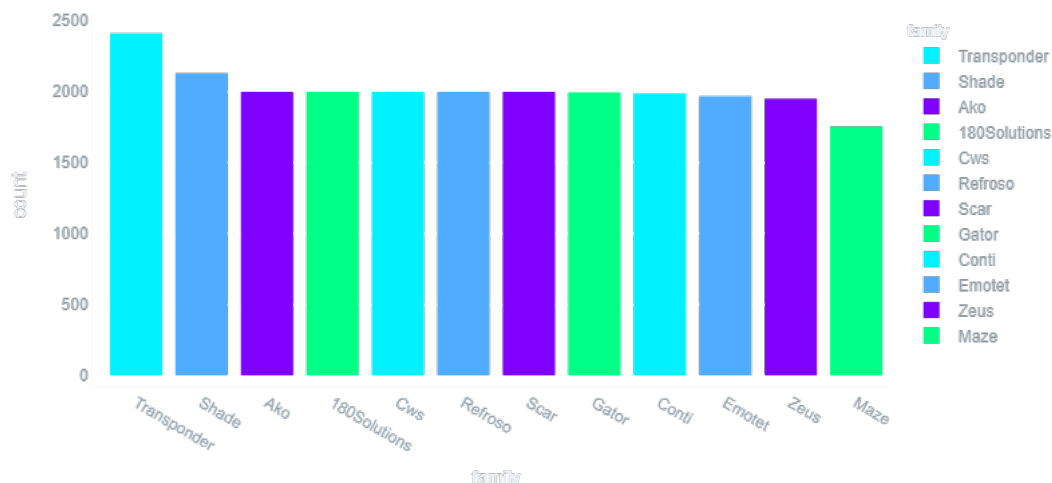


Figure 4.5: Malware family memory classification distribution.

4.7 Threat Attribution and Resolution Efficiency

- **Purpose:** To explore the attribution of cyber threats (hackers, nation-states, insiders) and correlate attack vectors with operational resolution efficiency.
- **What it represents:** An interactive strip/scatter plot plotting incident resolution time (hours) against defense strategy (VPN, Encryption, Antivirus, AI-based Detection, Firewall), with each point colored by attack vector (Phishing, Ransomware, Man-in-the-Middle, DDoS, SQL Injection, Malware).
- **Resolution Times:** Resolution times indicate how different defense mechanisms impact incident lifecycle lengths across threat types. Across all five defense strategies, resolution times are broadly distributed from near-immediate remediation up to roughly 70 hours, with no single mechanism showing a clearly compressed or superior resolution profile in isolation.
- **Attribution Share:** Nation-state groups and advanced insider threats typically represent distinct risk profiles in financial damage per security event, even when the raw resolution-time distributions across defense mechanisms appear similar — underscoring that resolution speed alone does not capture the full severity of an attribution class.
- **Business Interpretation:** The dense overlap of attack vectors across every defense strategy suggests that no single control (VPN, Encryption, Antivirus, AI-based Detection, or Firewall) is uniquely effective against a specific class of threat; layered, defense-in-depth postures remain necessary.
- **Key Takeaway:** Organizations should pair defense-mechanism selection with threat-attribution context (nation-state vs. insider vs. opportunistic) rather than relying on resolution-time averages alone to judge control effectiveness.

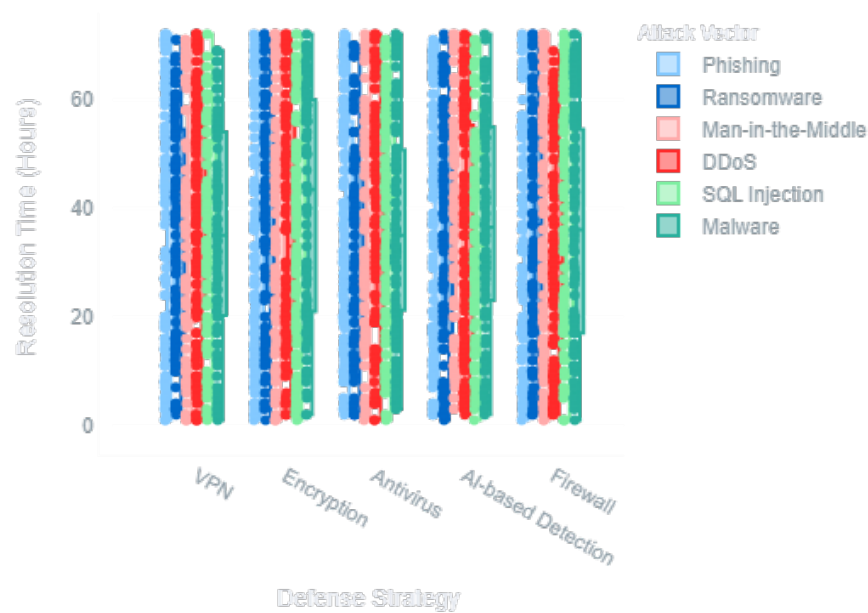


Figure 4.6: Resolution time (hours) by defense strategy, colored by attack vector.

4.8 Geopolitical Relationship Network

- **Purpose:** To visualize the flow of cyber threat operations.
- **What it represents:** A NetworkX force-directed graph connecting threat sources, target countries, and industries.
- **Observations:** Pink nodes (Threat Sources) like “Nation-state” and “Hacker Group” display high connectivity to primary sectors (Energy, Government, Banking) in specific countries (UK, USA, Germany).
- **Business Interpretation:** Advanced Persistent Threats (APTs) target critical national infrastructure to achieve geopolitical influence.
- **Key Takeaway:** National cyber defense requires public-private partnerships, as private companies manage the infrastructure targeted by nation-state actors.

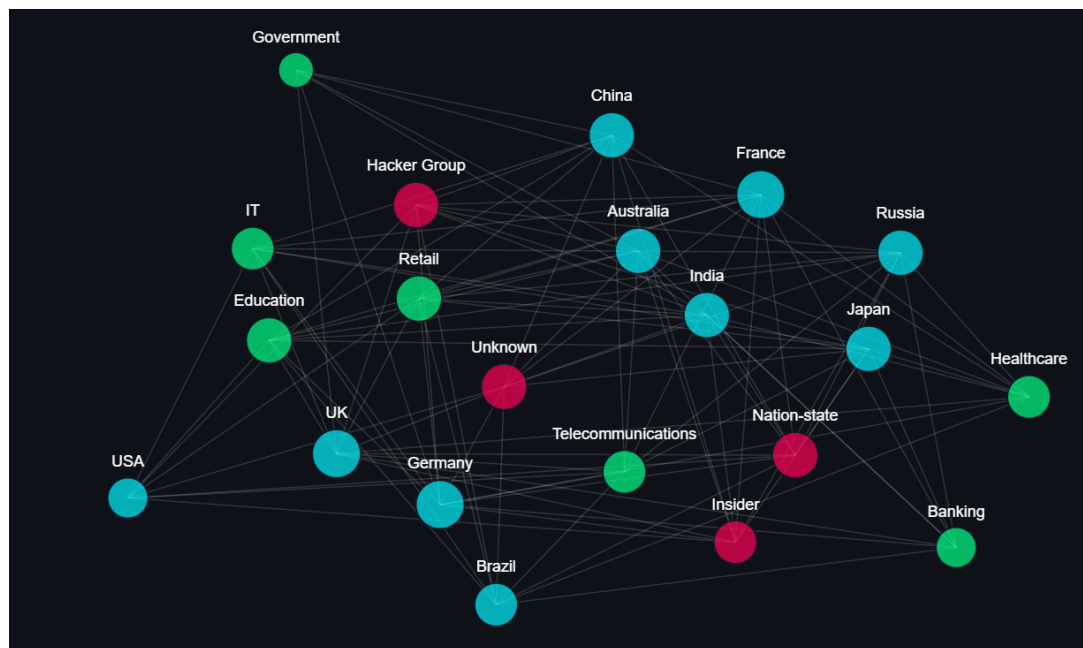


Figure 4.7: Threat actor relationship network mapping connections between threat vectors and sectors.

Chapter 5

Results and Findings

CyberVision provides structured, quantitative findings derived directly from the integrated data.

5.1 Country Risk Leaderboard

Aggregated results from `integrated_spatial.csv` show the national risk indices calculated under standard weights ($w_{\text{freq}} = 0.3$, $w_{\text{loss}} = 0.4$, $w_{\text{time}} = 0.3$).

Table 5.1: National Threat Risk Leaderboard

Rank	Country	Incidents	Total Loss (\$M)	Avg. Resolution (hrs)	Risk Score
1	United Kingdom	321	15,746.49	37.1	7.68
2	Brazil	310	16,880.70	36.8	6.97
3	Germany	291	14,767.36	37.5	5.58
4	Japan	305	14,871.81	36.4	5.31
5	Australia	297	15,403.00	36.5	5.18
6	France	305	15,622.91	36.0	4.17
7	India	308	14,917.87	35.8	3.21
8	Russia	295	15,219.18	35.7	3.11
9	United States	287	14,420.09	36.2	3.07
10	China	281	13,958.09	35.8	2.57

5.2 Industry Sector Loss Projections

Summarized sector distributions expose which industries bear the largest financial burdens:

- **IT Infrastructure:** Registers the highest average financial loss per incident (\$51.2M), indicating that IT services are highly leveraged targets.
- **Healthcare:** Records the largest aggregate financial loss (\$21,804M), reflecting high compliance and remediation costs.
- **Banking & Financial Services:** Follows closely with a total loss of \$21,120M, driven by direct theft targets.

5.3 Network Traffic Protocol Share

From the analysis of 40,000 traffic records in `attack_signatures_clean.csv`, the distribution of network protocols is:

- **ICMP (Internet Control Message Protocol):** 13,429 records (33.57% share) — typically associated with ping sweeps and denial-of-service amplification.

- **UDP (User Datagram Protocol):** 13,299 records (33.25% share) — indicating high-volume traffic flooding attacks.
- **TCP (Transmission Control Protocol):** 13,272 records (33.18% share) — representing standard connection-oriented exploit payloads.

5.4 Vulnerability Severity Breakdown

The distribution of the 3,111 software vulnerabilities published between 2021 and 2023 is:

- **Moderate Severity:** 1,447 CVEs (46.51%)
- **High Severity:** 1,048 CVEs (33.68%)
- **Critical Severity:** 449 CVEs (14.43%)
- **Low Severity:** 167 CVEs (5.38%)

Chapter 6

Discussion

6.1 Geopolitical and Economic Risk Dynamics

The country risk leaderboard indicates that cyber risk is not a direct reflection of a country's size or population. While the United Kingdom and Brazil occupy the top ranks, the United States and China exhibit lower scores. This pattern is driven by the components of the Country Risk Index:

- Brazil exhibits the highest total financial loss (\$16,880.70M), indicating a lack of robust security controls.
- The UK reports the highest number of incidents (321) and a high average resolution time (37.1 hours), reflecting active targeting and regulatory reporting requirements.
- The US and China exhibit lower average resolution times and lower reported average financial losses per incident in this dataset, which drags down their normalized risk scores.

It is worth stressing that this ranking is a function of the *reported* data and the chosen weighting scheme, not an absolute measure of national cybersecurity maturity; a jurisdiction with stricter incident-disclosure requirements will mechanically report more incidents than one without such requirements, independent of its true underlying exposure.

6.2 Vulnerability Lifecycle and Threat Landscape Shift

The CVE severity distribution highlights a shift in threat dynamics. The concentration of High and Critical vulnerabilities in 2023 suggests that software development lifecycles struggle to secure complex, cloud-native deployments. The lag in patch deployment allows threat actors to exploit vulnerabilities before organizations can apply updates, which is consistent with the broader industry observation that the median time-to-patch for critical vulnerabilities routinely exceeds the time-to-exploit once a proof-of-concept is public.

6.3 Practical Significance and Cyber Defense Tactics

The findings of CyberVision have practical implications for security operations:

1. **Dynamic Risk Alignment:** Organizations must calibrate security budgets to match their risk profiles (e.g. financial damage vs. operational uptime).
2. **Memory Forensic Deployment:** Endpoint detection tools must incorporate machine learning models capable of analyzing high-dimensional memory structures to identify obfuscated malware processes.
3. **Deep Packet Filtering:** Firewalls must inspect large packets, which are strongly correlated with anomalous traffic signatures.

6.4 Limitations of the Analysis

- **Static Weights:** Although user-adjustable in the UI, default weights in the backend code are hardcoded.
- **Geographic Representation:** Standardization maps geographic data to ten standard industrial nations, which limits the analysis of emerging markets.
- **Volatile Features:** The memory forensic dataset uses features specific to Volatility 2, which may not capture memory structures in modern operating systems.

Chapter 7

Challenges

7.1 High-Dimensional Volatility Memory Forensics

The MalMem dataset presents a classic “curse of dimensionality” challenge, containing 58 features representing process list states, loaded DLLs, and open handles. Visualizing these features in a 2D dashboard requires aggregation or dimensionality reduction (e.g. PCA), which can mask subtle indicators of malicious activity. Selecting which handful of derived dimensions to surface on a dashboard, without discarding the signal that a forensic analyst would actually need, proved to be one of the more delicate design trade-offs in the project.

7.2 Data Inconsistencies and Missing Value Imputations

Ingesting multiple CTI datasets introduces inconsistencies:

- Timestamps use varying formats, requiring robust date parsing routines.
- Missing categorical attributes require imputation to prevent data loss. Filling missing fields with “Unknown” preserves sample sizes but can introduce noise into categorical aggregations.

7.3 Network Graph Scaling and Layout Constraints

Visualizing the Geopolitical Relationship Network using NetworkX and Plotly introduces layout challenges. Projects with over 150 nodes suffer from overlapping nodes and visual clutter, reducing the legibility of threat actor paths. The static spring-layout algorithm lacks interactive physics, making it difficult to trace complex networks, and was ultimately one of the motivating factors behind the physics-based rendering proposed in Section 9.3.

Chapter 8

Conclusion

8.1 Summary of Accomplished Objectives

The development of CyberVision achieved its core goals:

- Designed and verified a reproducible data cleaning and integration pipeline.
- Developed a dynamic Country and Industry Risk Index using Min–Max scaling.
- Implemented high-contrast, neon-colored visualizations to analyze threat trends, network forensics, and memory states.
- Created a cached, multi-page Streamlit dashboard featuring a modern glassmorphic theme.

8.2 Key Analytical Insights

- Peak attack periods correlate with global security events (e.g. ransomware outbreaks in 2017, geopolitical tensions in 2022).
- Highly digitized, service-oriented economies exhibit higher reported risk indices.
- Network traffic analysis shows a strong correlation between large packet lengths and high anomaly scores, indicating payload-based exploits.

8.3 Platform Success and Practical Utility

CyberVision demonstrates the value of interactive big data visualization in security operations. By unifying disparate telemetry feeds, the platform reduces cognitive load and alert fatigue, enabling security analysts to make data-driven defensive decisions rather than manually reconciling isolated tools and spreadsheets.

Chapter 9

Future Scope

9.1 Predictive Alerting Engines and Time-Series Forecasting

A logical next step for CyberVision is the integration of predictive engines. Incorporating forecasting models (e.g. Prophet or ARIMA) would enable the platform to project future incident volumes based on historical trends, helping organizations allocate security resources proactively rather than reactively.

9.2 Integration of Standard CVSS Severity in Country Risk Calculations

To improve the Country Risk Index, future versions should incorporate CVSS (Common Vulnerability Scoring System) severity levels. Weighting incidents by vulnerability severity would provide a more accurate representation of risk than simple incident counts, since two countries with identical incident counts may face very different underlying exposure if one set of incidents involves critical remote-code-execution vulnerabilities and the other does not.

9.3 Interactive Physics-Based Graph Renderings

Replacing static NetworkX/Plotly network graphs with physics-based, interactive libraries (e.g. Pyvis or D3.js) would enhance usability. Dynamic layouts would resolve node overlap and allow analysts to manipulate nodes to inspect threat paths, directly addressing the scaling constraints identified in Section 7.3.

9.4 Cloud Deployment and Real-Time Stream Integration

Deploying CyberVision to cloud platforms (e.g., AWS, GCP) and integrating real-time message brokers (e.g. Apache Kafka) would transform the dashboard from a batch-oriented analytics tool into a real-time monitoring system, capable of ingesting and visualizing telemetry as it arrives rather than on a periodic refresh cycle.

Chapter 10

Code and Deployment Links

The full source code, preprocessing pipelines, and Streamlit application described in this report are publicly available, and a live instance of the dashboard is deployed for direct interaction without any local setup.

- **GitHub Repository (Group):** github.com/haryashva/Cyber-Attach-Visualisation
- **Live Deployed Web Application:** cyber-attach-visualisation.streamlit.app

Readers are encouraged to interact with the live deployment directly while reading Chapter 4, since the Risk Index Calibration sliders and tab-based navigation described in Section 4.1 are best understood interactively rather than through static screenshots alone.

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4. Hagberg, A. A., Schult, D. A., & Swart, P. J. (2008). *Exploring network structure, dynamics, and function using NetworkX*. Proceedings of the 7th Python in Science Conference.
5. Canadian Institute for Cybersecurity (2022). *CIC-MalMem-2022: Obfuscated Malware Memory Dataset*. University of New Brunswick. <https://www.unb.ca/cic/datasets/malmem-2022.html>
6. Kaggle API Documentation (2024). *Programmatic access to Kaggle datasets and competitions*. <https://github.com/Kaggle/kaggle-api>
7. National Vulnerability Database (NVD) (2024). *Common Vulnerabilities and Exposures (CVE) database*. National Institute of Standards and Technology. <https://nvd.nist.gov>

Appendix A

Appendices

A.1 Appendix A: Complete Repository Folder Structure

```
1 cybervision/
2 |-- app.py                # Entry point for layout and page routing
3 |-- requirements.txt      # Environment dependencies
4 |-- runtime.txt          # Python runtime environment version
5 |-- .env                 # Environment configuration
6 |-- backend/             # Business logic and analytical models
7 |   |-- loader.py        # Data loader and cache routines
8 |   |-- integration.py    # Country and attack type mapping
9 |   |-- risk.py           # Country and industry risk engines
10 |   |-- filters.py        # Sidebar widget query executors
11 |   |-- analytics.py      # Metric card counts and sums
12 |-- frontend/           # Presentation layers and UI styles
13 |   |-- layout.py        # CSS injections and dark neon theme config
14 |   |-- sidebar.py       # Navigation and sidebar sliders
15 |   |-- components.py    # Interactive Plotly chart builders
16 |-- pages/              # Individual page modules
17 |   |-- Home.py          # High-level overview and KPIs
18 |   |-- Global_Threats.py # Annual threat trends
19 |   |-- Country_Map.py   # World choropleth map
20 |   |-- Industry.py      # Industry Treemap and leaderboard
21 |   |-- Vulnerability.py # CVE search list and heatmap
22 |   |-- Attack_Source.py # Traffic anomalies scatter
23 |   |-- Network.py       # Threat actor networks
24 |-- data/               # Data storage layers (gitignored)
25 |   |-- raw/             # Raw CSV datasets
26 |   |-- interim/         # Intermediary clean states
27 |   |-- processed/       # Preprocessed CSVs used by app
28 |-- src/                # Data engineering source code
29 |   |-- preprocessing/   # Cleaning pipelines
30 |   |   |-- base_cleaning.py # Reusable cleaning routines
31 |   |   |-- clean_*.py      # Dataset-specific cleaning scripts
32 |   |   |-- integrate_data.py # Spatial and temporal aggregations
33 |   |   |-- run_all.py      # Automation script
34 |   |-- utils/           # Utilities folder
35 |-- docs/               # Technical reports and documentation
```

A.2 Appendix B: Software and Package Requirements

The system requires Python 3.9+ and the packages listed in `requirements.txt`:

```
1 pandas >=2.2
2 numpy >=1.26
3 python-dateutil >=2.9
4 plotly >=5.22
5 matplotlib >=3.9
6 networkx >=3.3
7 streamlit >=1.36
8 geopandas >=0.14
9 pycountry >=23.12
10 scikit-learn >=1.5
11 prophet >=1.1
12 kaggle >=1.6
```

```

13 jupyter>=1.0
14 ipykernel>=6.29
15 pytest>=8.2
16 black>=24.4
17 python-dotenv>=1.0

```

A.3 Appendix C: Environment Setup and Execution Guide

To deploy CyberVision locally:

1. Clone the repository:

```

1 git clone <repo-url>
2 cd cybervision

```

2. Create and activate a virtual environment:

```

1 python3 -m venv venv
2 # Windows
3 venv\Scripts\activate
4 # Linux/macOS
5 source venv/bin/activate

```

3. Install dependencies:

```

1 pip install -r requirements.txt

```

4. Acquire the datasets:

- Set up Kaggle API credentials (~/.kaggle/kaggle.json).
- Run the automated download script:

```

1 python -m src.preprocessing.download_data

```

- Manually download cic_malmem_2022.csv from the UNB portal and place it in data/raw/.

5. Run the cleaning and integration pipelines:

```

1 python -m src.preprocessing.run_all

```

6. Launch the Streamlit web application:

```

1 streamlit run app.py

```

Open <http://localhost:8501> in your web browser.

A.4 Appendix D: Important Preprocessing & Risk Logic Snippets

Cleaning Pipeline (Derived from base_cleaning.py)

```

1 def standardize_column_names(df: pd.DataFrame) -> pd.DataFrame:
2     df = df.copy()
3     df.columns = (
4         df.columns.str.strip()
5         .str.lower()
6         .str.replace(r"\s+", "_", regex=True)
7         .str.replace(r"[^\w]", "", regex=True)
8     )
9     return df
10
11 def fill_missing(df: pd.DataFrame, strategy: dict) -> pd.DataFrame:
12     df = df.copy()
13     for col, method in strategy.items():
14         if col not in df.columns:
15             continue
16         if method == "mean":
17             df[col] = df[col].fillna(df[col].mean())
18         elif method == "median":
19             df[col] = df[col].fillna(df[col].median())
20         elif method == "mode":
21             df[col] = df[col].fillna(df[col].mode().iloc[0])
22         else:
23             df[col] = df[col].fillna(method)
24     return df

```

Custom Risk-Scoring Engine (Derived from backend/risk.py)

```

1 def calculate_normalized_scores(
2     df: pd.DataFrame,
3     group_col: str,
4     weight_count: float = 0.3,
5     weight_loss: float = 0.4,
6     weight_time: float = 0.3
7 ) -> pd.DataFrame:
8     if df.empty or group_col not in df.columns:
9         return pd.DataFrame(columns=[group_col, "incident_count",
10                                     "avg_loss", "avg_resolution_time", "risk_score"])
11
12     grouped = df.groupby(group_col)
13     counts = grouped.size().reset_index(name="incident_count")
14
15     loss_col = "financial_loss_in_million_"
16     losses = grouped[loss_col].mean().reset_index(name="avg_loss") \
17         if loss_col in df.columns else \
18         pd.DataFrame({group_col: counts[group_col], "avg_loss": 0.0})
19
20     time_col = "incident_resolution_time_in_hours"
21     times = grouped[time_col].mean().reset_index(name="avg_resolution_time") \
22         if time_col in df.columns else \
23         pd.DataFrame({group_col: counts[group_col], "avg_resolution_time":
24                        0.0})
25
26     metrics = pd.merge(counts, losses, on=group_col)
27     metrics = pd.merge(metrics, times, on=group_col)
28
29     def minmax_scale(series: pd.Series) -> pd.Series:
30         min_val = series.min()
31         max_val = series.max()
32         if max_val == min_val:
33             return pd.Series(0.5, index=series.index)
34         return (series - min_val) / (max_val - min_val)

```

```
34
35     norm_count = minmax_scale(metrics["incident_count"])
36     norm_loss = minmax_scale(metrics["avg_loss"])
37     norm_time = minmax_scale(metrics["avg_resolution_time"])
38
39     weighted_score = (weight_count * norm_count + weight_loss * norm_loss
40                       + weight_time * norm_time) * 10.0
41     metrics["risk_score"] = np.round(weighted_score, 2)
42     return metrics.sort_values(by="risk_score", ascending=False)
```